Himanshu Kumar

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IIT Bhubaneswar

Signal Processing and Communication Engg

M.Tech - 2024–26

Computer Vision/Machine Learning

Education

Year	Degree	University/Board	%/CGPA
2025	M.Tech Exchange	**University Of Agder , Norway**	
2024-26	M.Tech in Signal Processing and Communication	**IIT , Bhubaneswar**	**7.17/10**
2020-24	B.Tech in Mechatronics Engineering	**Puducherry Technological University**	**8.56/10**
2020-24	Minors in Computer Science and Engineering	**Puducherry Technological University**	**7.00/10**
2018	Intermediate (12^{th} standard)	**CBSE**	**87.6%**
2016	Secondary School Certificate (10^{th})	**CBSE**	**10/10**

Publications

- **ICCAI 2026:** A Customized Lightweight CNN for RF Spectrogram–Based Multi-Class Drone Classification
  - Engineered a lightweight CNN architecture using STFT-based RF spectrogram processing for robust Non-Line-of-Sight (NLoS) UAV detection.
  - Achieved **94.90%** multi-class classification accuracy across **24 drone classes** on the Drone RFb-Spectra dataset.
  - Optimized the model to only **266.14 KB** storage size, enabling computationally efficient deployment for embedded surveillance and edge systems.
- **IC2E3 2026 (IEEE IAS):** Ghost-CAS-UNet for On-Device Diabetic Retinopathy Screening: Lightweight Retinal Lesion Segmentation on Edge Hardware
  - Developed a lightweight retinal lesion segmentation framework with only **1.98M parameters** and **7.88 MB** model size for edge deployment.
  - Designed Ghost-CAS-UNet integrating Ghost modules, Coordinate Attention, and attention-gated skip fusion, achieving **39.00% Mean Dice** and **30.24% Mean IoU** on the IDRiD dataset.
  - Implemented a memory-efficient sliding-window inference pipeline with bounded peak RAM of ∼**684 MB** on Raspberry Pi 4, while improving Microaneurysm detection (**15.97% Dice** vs **0.00%** baseline collapse).

Projects

Reinforcement Learning-based Hybrid Communication Protocol for FANETs — ***M.Tech Research Project***

- Designed and simulated UAV communication networks using TDMA/CSMA protocols, focusing on network performance under varying load, mobility, and channel conditions.
- Modeled realistic network environments including latency, packet loss, fading (AWGN, Rayleigh, Rician, Nakagami-m), and link reliability metrics (BER, path loss).
- Developed Python-based automation pipelines for running simulations, logging results, and performance evaluation.
- Implemented distributed system architecture (multi-agent framework) reflecting decentralized network control.
- **Tech Stack**: Python, Linux, NumPy/SciPy, PyTorch, Git

ANAV for SLAM and Path Planning — ***ISRO USRC Challenge 2025***

- Built and configured a Linux-based embedded system (Jetson Nano + Pixhawk 2 + ZED2 camera) for real-time autonomous navigation.
- Established inter-device communication pipeline using ROS2 over serial interfaces, enabling reliable data exchange across components.
- Performed system integration, debugging, and troubleshooting of hardware-software interfaces in real-time environments.
- Implemented fault-handling mechanisms (auto return on signal loss/low battery) improving system reliability.
- **Tech Stack**: Linux, ROS2, Python, Docker, Git, Embedded Systems, Networking Basics

Internship Experience

Bharat Space Education Research Centre (DEF-SPACE, supported by ISRO) — Dec 2025 - Jan 2026
Winter Intern (Aerospace & Robotics Modelling — MATLAB, Python, Git)

- Delivered : Tail-sitting **VTOL** concept, fixed-wing preliminary sizing, aero–stability, water-rocket trajectory, and **4-DOF** robot arm motion planning.
- Aircraft sizing: computed **W/S** and **T/W** trade-offs and estimated V_s and endurance-level feasibility from first-order performance relations.
- Aero–stability: evaluated **trim** behavior and quantified **CG sensitivity** effects on glide/flight response over operating-point sweeps.
- Robotics: implemented **forward + inverse kinematics** for a **4-DOF** manipulator and generated smooth joint-space trajectories for pick-and-place style motion.

Technical Skills

Programming & Scripting: Python, C++, SQL, Bash (Linux CLI), Basic PowerShell
Systems & Networking: Linux System Administration, TCP/IP Basics, Network Configuration,System Monitoring
Tools & Platforms: Git, Docker, AWS (Basic), Virtual Environments, ROS2
Relevant Coursework: Operating Systems, Computer Networks,Distributed Systems, System Programming, Wireless Communication

Workshops and Conferences

- GIAN Short Course @IIT BBS (Dec 2024): RISC-V, AI, TL-Verilog
- PCB Design using Altium Designer Workshop, 3D Modelling Printing
- VECTOR Solutions Workshop (Mar 2022)

Scholastic Achievements

- Qualified GATE ECE (2024 / 2025)
- Qualified UGC NET (2025)
- Qualified for Assistant System Engineer role at TCS (2024)

Positions of Responsibility

Teaching Assistant Jul 2024 – Present
Digital Electronics (Lab/Theory)/Satellite Communication (Theory) , IIT Bhubaneswar
ESN UiA — Erasmus Student Network, University of Agder (Norway) Volunteer (Exchange Semester)

- Supported community engagement programs with elderly residents through structured activities and peer coordination.
- Assisted event operations (setup, attendee flow, group coordination) with ESN mentors for cross-cultural integration.

Robotics Club of PTU Mar 2022 – Feb 2024
Design & Organizing Head

- Led *Tertonx Symposium* and technical event operations; owned design collateral, event flow, and judge/mentor coordination.
- Partnered with startups to drive participation and sponsorship discussions; managed outreach and coordination for event days.

MUN Society of PTU Jul 2023 – Feb 2024
Organizing Head

- Owned end-to-end delivery of Model UN & Indian Parliament simulations: agenda design, delegate operations, vendor/logistics, and on-ground execution.
- Led flagship sessions for *CURIOSO* (Literary Club), coordinating committees, timelines, and cross-team workflows.

Hobbies and Interests

- Vibe coding | MUN & debating | travelling and connecting with people worldwide | singing | weightlifting